

LOEI AGROMET, Thailand

WMO: 483500

Lat: 17.4094N

Lon: 101.7294E

Elev: 865

StdP: 14.24

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	53.0	56.5	48.6	52.2	59.4	51.6	58.5	60.6	9.5	79.5	8.3	78.8	0.2	N/A	0.261

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	22.0	100.4	78.7	97.7	78.1	95.3	78.4	82.3	92.3	81.6	91.3	81.0	90.3	4.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.8	159.5	88.1	79.1	155.8	87.3	78.5	152.8	86.3	47.0	92.5	46.3	91.9	45.6	91.2	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.4	6.4	DB	47.2	102.9	4.8	2.8	43.7	105.0	40.9	106.6	38.1	108.2	34.6	110.3	(4)
(5)			WB	47.0	84.1	4.6	1.0	43.7	84.8	41.0	85.4	38.5	86.0	35.1	86.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	79.5	72.3	76.6	80.4	84.3	83.2	83.4	82.4	81.7	81.4	79.7	76.9	72.2		
	DBStd	5.33	4.64	4.34	4.61	3.97	2.92	2.28	2.62	2.21	1.88	2.57	3.49	4.90		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	19	8	1	2	0	0	0	0	0	0	0	0	8		
	CDD50	10787	691	744	944	1030	1031	1001	1005	983	942	920	808	688		
	CDD65	5331	234	325	481	580	566	551	540	518	492	455	358	231		
	CDH74	49463	1690	3048	5043	6552	5511	5782	5295	4631	4150	3492	2640	1629		
	CDH80	20887	534	1418	2675	3595	2421	2388	1985	1628	1451	1309	980	503		
(14) Wind	WSAvg	1.4	1.4	1.7	1.7	1.7	1.3	1.3	1.3	1.2	1.1	1.1	1.2	1.3		
(15) Precipitation	PrecAvg	48.0	0.2	0.6	1.5	3.7	8.0	5.8	5.4	7.4	9.8	4.4	0.7	0.2		
	PrecMax	60.7	1.7	2.1	2.6	7.0	15.3	13.3	7.7	13.8	20.8	8.3	2.1	1.6		
	PrecMin	39.7	0.0	0.0	0.0	0.8	3.1	1.9	2.0	4.1	5.2	0.8	0.0	0.0		
	PrecStd	6.1	0.5	0.7	0.8	1.9	3.2	2.9	1.7	2.9	4.6	2.4	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.2	97.1	102.1	105.2	100.3	97.0	98.5	94.2	93.3	93.9	92.6	90.8		
		MCWB	73.4	74.6	77.3	79.7	81.3	80.6	79.8	81.1	81.8	80.1	78.7	75.3		
	2%	DB	87.9	94.3	98.9	101.9	96.6	94.3	94.4	91.8	91.5	91.0	89.7	88.0		
		MCWB	72.5	74.4	75.7	79.3	81.2	80.5	80.3	80.4	81.2	78.7	77.1	74.8		
	5%	DB	85.8	91.9	96.4	98.8	93.7	92.2	91.3	89.8	89.7	89.1	88.0	85.3		
		MCWB	72.2	73.9	75.7	78.8	80.7	80.4	80.0	79.9	80.6	78.6	76.6	73.6		
	10%	DB	83.2	88.9	93.0	95.7	91.0	90.0	88.6	88.0	87.8	87.1	86.0	82.5		
		MCWB	70.9	73.3	75.4	78.4	80.2	80.0	79.5	79.5	80.0	77.9	75.5	71.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.2	78.1	79.9	82.4	83.3	82.7	82.5	82.6	82.8	81.4	80.0	78.0		
		MCDB	86.1	91.6	96.7	99.0	95.2	92.1	91.7	91.0	90.8	89.7	89.1	87.1		
	2%	WB	74.9	76.3	78.2	81.0	81.9	81.7	81.3	81.3	81.8	80.2	78.7	75.8		
		MCDB	83.9	90.0	93.5	96.4	93.0	91.1	90.1	89.4	89.5	88.3	87.5	85.0		
	5%	WB	73.4	75.2	77.1	80.0	81.1	80.9	80.5	80.4	81.0	79.4	77.4	74.2		
		MCDB	81.8	88.8	91.9	94.7	91.5	89.9	89.0	88.1	88.2	87.1	85.8	82.8		
	10%	WB	71.9	74.0	76.1	79.0	80.2	80.2	79.8	79.7	80.1	78.4	76.3	72.8		
		MCDB	79.9	86.1	90.1	93.1	89.7	88.5	87.4	86.7	86.6	85.5	83.7	80.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.0	25.5	24.0	22.0	16.5	14.5	13.4	13.1	13.7	15.2	18.5	21.1		
		MCDBR	25.2	29.5	30.2	26.8	20.8	17.6	17.8	16.1	16.6	18.3	20.3	22.8		
	5% WB	MCWBR	11.2	11.4	10.8	8.9	7.4	6.2	6.8	6.5	7.4	7.6	8.9	10.3		
		MCWBR	20.5	25.5	25.2	23.1	18.3	15.7	15.0	14.7	15.1	16.2	18.0	20.8		
(40) Clear-Sky Solar Irradiance	taub		0.466	0.571	0.670	0.623	0.494	0.444	0.432	0.447	0.494	0.538	0.453	0.456		
		taud	2.110	1.852	1.660	1.809	2.191	2.358	2.387	2.324	2.178	2.011	2.224	2.157		
	Ebn at Noon		258	236	218	230	258	269	272	271	259	242	260	257		
		Edh at Noon	48	65	81	70	47	40	39	41	48	55	43	45		
(44) All-Sky Solar Radiation	RadAvg		1475	1615	1632	1749	1643	1544	1445	1400	1426	1491	1533	1430		
	RadStd		99	75	103	95	113	110	99	89	89	102	64	58		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+209	+209

Nomenclature: See separate page